# Stochastic Models and Valuation in Finance

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## 1 Some basics from probability theory

**Lemma 1.1** (Substitution lemma)**.** *Let $X, Y$ be random variables on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $X$ is independent and $Y$ is $\mathcal{H}$-measureable with $\mathcal{H} \subset \mathcal{F}$. Moreover, let $u : \Omega \times \Omega \to \mathcal{E}$ be some measureable function, then*

$$\mathbb{E}(u(X,Y)|\mathcal{H}) = v(Y), \text{ with } v(y) := \mathbb{E}(u(X,y)) \tag{1}$$

*Proof.* Assume that $u$ is given by a characteristic function of the form $u(x,y) := \mathbb{1}_{A\times B}(x,y) = \mathbb{1}_A(x)\mathbb{1}_B(y)$ For any $A \in \mathcal{F}$ and $B \in \mathcal{H}$ one has

$$\begin{aligned}\mathbb{E}(u(X,Y)|\mathcal{H}) &= \mathbb{E}(\mathbb{1}_{\{X\in A\}}\mathbb{1}_{\{Y\in B\}}|\mathcal{H}) = \mathbb{1}_{\{Y\in B\}}\mathbb{E}(\mathbb{1}_{\{X\in A\}}) \\ &= (\omega \mapsto \mathbb{E}(u(X,Y(\omega)))) = v(Y)\end{aligned} \tag{2}$$

with $v(y) := \mathbb{E}(u(X,y))$. □

**Definition 1.2** (The (non-central) $\chi^2$-distribution)**.** Let $X_i : \Omega \to \mathbb{R}$, $i = 1, \dots, n$ be stochastically independent normally distributed random variables such that $X_i \sim \mathcal{N}(\mu_i, 1)$. The random variable $V := \sum_{i=0}^n X_i^2$ is called to be non-centrally $\chi_n^2(\lambda)$-distributed, in symbol $V \sim \chi_n^2(\lambda)$ with $n$ degrees of freedom and non-centrality parameter $\lambda := \sum_{i=0}^n \mu_i^2$.

**Lemma 1.3.** *Let $S^{n-1}$ for $n \in \mathbb{N}$ be the $n$-sphere in $\mathbb{R}^n$. The following integral identity holds*

$$\int_{S^{n-1}} \cos^{2m}\theta_1 \,\mathrm{d}\Omega = 2\pi^{\frac{n-1}{2}} \frac{\Gamma(m+\frac{1}{2})}{\Gamma(\frac{n}{2}+m)} \tag{3}$$

*Proof.* We consider the integral

$$I := \int_{\mathbb{R}^n} x_1^{2m} e^{-\pi\|x\|^2} \,\mathrm{d}^n x \tag{4}$$

Using the standard identity for Gaussian integrals this yields

$$\begin{aligned} I &= \int_{\mathbb{R}} x_1^{2m} e^{-\pi x_1^2} \,\mathrm{d}x_1 \left(\int_{\mathbb{R}} e^{-\pi x^2} \,\mathrm{d}x\right)^{n-1} \\ &= \int_{\mathbb{R}} x_1^{2m} e^{-\pi x_1^2} \,\mathrm{d}x_1 \\ &= \frac{1}{\pi^{m+\frac{1}{2}}} \int_0^\infty y^{m-\frac{1}{2}} e^{-y} \,\mathrm{d}y \\ &= \frac{1}{\pi^{m+\frac{1}{2}}} \Gamma\left(m+\frac{1}{2}\right) \end{aligned} \tag{5}$$

by definition of the Gamma function. On the hand, using spherical coordinates, one finds

$$\begin{aligned} I &= \int_0^\infty \int_{S^{n-1}} (r\cos\theta_1)^{2m} r^{n-1} e^{-\pi r^2} \,\mathrm{d}r\mathrm{d}\Omega \\ &= \int_0^\infty r^{2m+n-1} e^{-\pi r^2} \,\mathrm{d}r \int_{S^{n-1}} \cos^{2m}\theta_1 \,\mathrm{d}\Omega \\ &= \frac{1}{2\pi^{m+\frac{n}{2}}} \int_0^\infty y^{m+\frac{n}{2}-1} e^{-y} \,\mathrm{d}y \int_{S^{n-1}} \cos^{2m}\theta_1 \,\mathrm{d}\Omega \\ &= \frac{\Gamma\left(\frac{n}{2}+m\right)}{2\pi^{m+\frac{n}{2}}} \int_{S^{n-1}} \cos^{2m}\theta_1 \,\mathrm{d}\Omega \end{aligned} \tag{6}$$

Comparing both expressions and solving for the remaining surface integral this directly leads to ??. □

**Corollary 1.4.** *The pdf of the $\chi_n^2(\lambda)$ distribution is given by*

$$f_{n,\lambda}(v) = \frac{1}{2^{\frac{n}{2}}} v^{\frac{n}{2}-1} e^{-\frac{1}{2}(v+\lambda)} \sum_{k=0}^\infty \left(\frac{\lambda}{4}\right)^k \frac{v^k}{k!\Gamma\left(\frac{n}{2}+k\right)} \tag{7}$$

*Proof.* Since the $X_i$ for $i = 1, \ldots, n$ are mutually stochastically independent, the joint distribution is given by

$$\mathbb{P}(X_1 \le x_1, \ldots, X_n \le x_n) = \prod_{i=1}^n \mathbb{P}(X_i \le x_i) = \frac{1}{(2\pi)^{\frac{n}{2}}} \prod_{i=1}^n \int_{-\infty}^{x_i} e^{-\frac{1}{2}(y_i-\mu_i)^2} \,\mathrm{d}y_i \tag{8}$$

Hence, from this it follows

$$\begin{aligned} \mathbb{P}(V \le x) &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \mathbb{1}_{\{\sum_i x_i^2 \le x\}} e^{-\frac{1}{2}\sum_{i=1}^n (x_i-\mu_i)^2} \,\mathrm{d}^n x \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2}\lambda} \int_{B^n(x)} e^{-\frac{1}{2}\|x\|^2 + x^T\mu} \,\mathrm{d}^n x \end{aligned} \tag{9}$$

Let $A \in O(n)$ be an orthogonal transformation such that $A\mu = (\sqrt{\lambda}, 0, \ldots, 0)^T$, then, due to invariance of the Lebesgue integral under orthogonal transformations, we find

$$\begin{aligned} \mathbb{P}(V \le x) &= \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2}\lambda} \int_{B^n(\sqrt{x})} e^{-\frac{1}{2}\|A^T x\|^2 + (A^T x)^T \mu} \,\mathrm{d}^n x \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2}\lambda} \int_{B^n(\sqrt{x})} e^{-\frac{1}{2}\|x\|^2 + x^T(A\mu)} \,\mathrm{d}^n x \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2}\lambda} \int_{B^n(\sqrt{x})} e^{-\frac{1}{2}\|x\|^2 + \sqrt{\lambda} x_1} \,\mathrm{d}^n x \end{aligned} \tag{10}$$

Going over to spherical coordinates, this can be written as

$$\mathbb{P}(V \le x) = \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2}\lambda} \int_0^{\sqrt{x}} r^{n-1} e^{-\frac{1}{2}r^2} \left( \int_{S^{n-1}} e^{\sqrt{\lambda} r \cos\theta_1} \,\mathrm{d}\Omega \right) \mathrm{d}r \tag{11}$$

$$\int_{S^{n-1}} e^{\sqrt{\lambda} r \cos\theta_1} \,\mathrm{d}\Omega = \sum_{k=0}^\infty \frac{\lambda^k}{(2k)!} r^{2k} \int_{S^{n-1}} \cos^{2k}\theta_1 \,\mathrm{d}\Omega \tag{12}$$

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Performing the change of variables $v = \sqrt{x}$, this yields

$$f_{n,\lambda}(v) := \frac{d\mathbb{P}(V \leq v^2)}{dv} = \frac{1}{2^{\frac{n}{2}}} v^{\frac{n}{2}-1} e^{-\frac{1}{2}(v+\lambda)} \sum_{k=0}^{\infty} \left(\frac{\lambda}{4}\right)^k \frac{v^k}{k!\Gamma\left(\frac{n}{2}+m\right)} \tag{13}$$

□

**Corollary 1.5.** *The moment generating function (mgf) of the non-central $\chi_n^2(\lambda)$-distribution is given by*

$$M_{\chi_n^2(\lambda)}(t) = \frac{e^{-\frac{t}{2t+1}\lambda}}{(2t+1)^{\frac{n}{2}}} \tag{14}$$

*Proof.*

$$\begin{aligned}
M_{\chi_n^2(\lambda)}(t) &= \mathbb{E}\left(e^{-tV}\right) = \int_0^\infty e^{-tv} f_{n,\lambda}(v)\,\mathrm{d}v \\
&= \frac{1}{2^{\frac{n}{2}}} \int_0^\infty v^{\frac{n}{2}-1} e^{-\frac{1}{2}((2t+1)v+\lambda)} \sum_{k=0}^{\infty} \left(\frac{\lambda}{4}\right)^k \frac{v^k}{k!\Gamma\left(\frac{n}{2}+m\right)}\,\mathrm{d}v \\
&= \frac{1}{(2t+1)^{\frac{n}{2}}} \frac{1}{2^{\frac{n}{2}}} \int_0^\infty w^{\frac{n}{2}-1} e^{-\frac{1}{2}(w+\frac{\lambda}{2t+1}(2t+1))} \sum_{k=0}^{\infty} \left(\frac{\lambda}{4(2t+1)}\right)^k \frac{w^k}{k!\Gamma\left(\frac{n}{2}+m\right)}\,\mathrm{d}w \\
&= \frac{e^{-\frac{\lambda t}{2t+1}}}{(2t+1)^{\frac{n}{2}}} \int_0^\infty f_{n,\frac{\lambda}{2t+1}}(w)\,\mathrm{d}w \\
&= \frac{e^{-\frac{t}{2t+1}\lambda}}{(2t+1)^{\frac{n}{2}}}
\end{aligned} \tag{15}$$

□

**Corollary 1.6.** *The mean and variance of the non-central $\chi_n^2(\lambda)$-distribution are given by*

$$\mathbb{E}(V) = n + \lambda, \quad \mathbb{V}ar(V) = 2(n+2\lambda) \tag{16}$$

*Proof.*

$$\begin{aligned}
\mathbb{E}(V) &= -\frac{d}{dt}\bigg|_{t=0} M_{\chi_n^2(\lambda)}(t) = \left[\frac{e^{-\frac{t}{2t+1}\lambda}}{(2t+1)^{\frac{n+4}{2}}}\left((2t+1)n+\lambda\right)\right]\bigg|_{t=0} \\
&= \left[M_{\chi_{n+4}^2(\lambda)}\left((2t+1)n+\lambda\right)\right]\bigg|_{t=0} = \lambda + n
\end{aligned} \tag{17}$$

$$\begin{aligned}
\mathbb{E}(V^2) &= \frac{d^2}{dt^2}\bigg|_{t=0} M_{\chi_n^2(\lambda)}(t) = -\frac{d}{dt}\bigg|_{t=0} \left[M_{\chi_{n+4}^2(\lambda)}\left((2t+1)n+\lambda\right)\right] \\
&= (n+4+\lambda)(n+\lambda) - 2n = (n+\lambda)^2 + 2(n+2\lambda)
\end{aligned} \tag{18}$$

Hence, using $\mathbb{V}ar(V) = \mathbb{E}(V^2) - (\mathbb{E}(V))^2$ this yields ??. □

Formula (1.4) remains well-defined if we replace $n \in \mathbb{N}$ by an arbitrary real parameter $d > 0$. This motivates the following definition.

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**Definition 1.7** (Generalized non-central $\chi^2$ distribution)**.** Let $d > 0$ and $\lambda \geq 0$. A random variable $V$ is said to follow a generalized non-central chi-square distribution with $d$ degrees of freedom and non-centrality parameter $\lambda$, written $V \sim \chi_d^2(\lambda)$, if its density is given by

$$f_{d,\lambda}(v) = \frac{1}{2} e^{-(v+\lambda)/2} \sum_{k=0}^{\infty} \frac{(\lambda/2)^k}{k!} \frac{v^{\frac{d}{2}+k-1}}{2^{\frac{d}{2}+k}\Gamma\left(\frac{d}{2}+k\right)}, \qquad v > 0. \tag{19}$$

For $\lambda = 0$, this reduces to the Gamma distribution $\Gamma(d/2, 2)$, i.e. the central chi-square distribution with $d$ degrees of freedom.

# 2 Monte Carlo simulation

**Definition 2.1** (Weak law of large numbers)**.** Let $(X_i)_{i\geq 1}$ be a sequence of random variables and define

$$S_n := \frac{1}{n}\left(\sum_{i=1}^{n} X_i - \mathbb{E}(X_1)\right).$$

We say that the weak law of large numbers (WLLN) holds if

$$\forall \varepsilon > 0 : \quad \mathbb{P}\left(|S_n| > \varepsilon\right) \xrightarrow[n\to\infty]{} 0. \tag{20}$$

**Definition 2.2** (Strong law of large numbers)**.** Let $(X_i)_{i\geq 1}$ be a sequence of random variables and define

$$S_n := \frac{1}{n}\left(\sum_{i=1}^{n} X_i - \mathbb{E}(X_1)\right).$$

We say that the strong law of large numbers (SLLN) holds if

$$\mathbb{P}\left(\lim_{n\to\infty} S_n = 0\right) = 1. \tag{21}$$

**Theorem 2.3** (Weak law under finite variance)**.** *Let $(X_i)_{i\geq 1}$ be independent and identically distributed random variables with $\mathbb{E}[X_1] = \mu$ and $\mathrm{Var}(X_1) < \infty$. Then the weak law of large numbers holds.*

**Theorem 2.4** (Strong law under finite expectation)**.** *Let $(X_i)_{i\geq 1}$ be independent and identically distributed random variables with $\mathbb{E}[X_1] = \mu$ and $\mathbb{E}[|X_1|] < \infty$. Then the strong law of large numbers holds.*

**Definition 2.5** (The cumulative distribution function)**.** Let $X : \Omega \to \mathbb{R}$ be a real random variable. The *cumulative distribution function (CDF)* of $X$, denoted by $F_X : \mathbb{R} \to [0,1]$, is defined as

$$F_X(x) := \mathbb{P}(\{X \leq x\}) \tag{22}$$

Since $\mathbb{P}(\{X \leq x\}) = \mathbb{E}(\mathbb{1}_{\{X\leq x\}})$, using the pushforward measure w.r.t X, this can also be written as

$$F_X(x) = \int_{-\infty}^{x} \lambda \, d(X_*\mathbb{P}) \tag{23}$$

**Example 2.6.** For a family of independent and identically distributed real random variables $(X_i)_i$ for $i = 1, \ldots, n$ with $X_i \sim X$, $\forall i = 1, \ldots, n$, for some real random variable $X$ consider $Y_i(x) := \mathbb{1}_{\{X_i \leq x\}} : \Omega \to \mathbb{R}$. Then, the $Y_i$ also defines a i.i.d family of real random variables which, by the strong law of large numbers, satisfies

$$\frac{1}{n}\sum_{i=1}^{n} Y_i(x) \xrightarrow{n\to\infty} \mathbb{E}(\mathbb{1}_{\{X\leq x\}}) = F_X(x), \quad \forall x \in \mathbb{R} \tag{24}$$

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**Example 2.7** (The Kolmogorov-Smirnov test)**.** As a typical example, consider the drawing of $n \in \mathbb{N}$ random samples $x_i := X_i(\omega)$, $i = 1, \dots, n$, out of a box and counting the total number of subsammples $x_j$, $j \in J$ with $x_j \leq x$, i.e. considering the number $|\{x_i | \, x_i \leq x\}|$. Then, in the limit of growing the sample size to infinity, the relative fraction of those samples which are less or equal to $x$ converges against the CDF of $X$, i.e.,

$$\frac{|\{x_i | \, x_i \leq x\}|}{n} \xrightarrow{n\to\infty} F_X(x) \tag{25}$$

**Example 2.8** (The Q-Q plot)**.**

$$F^{-1}(F_n(x)) \xrightarrow{n\to\infty} x \tag{26}$$

**Example 2.9** (Monte Carlo Simulation)**.** Let $(X_i)_i$ for $i = 1, \dots n$ be a family of independent and identically distributed random variables such that $X_i \sim \mathbb{Q}$ $\forall i$, then

$$\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}(f(X_i)) = \frac{1}{n}\sum_{i=1}^{n} \int_\Omega f(X_i)\, d\mathbb{P} = \frac{1}{n}\sum_{i=1}^{n} \int_\mathbb{R} f(x)\, d(X_{i*}\mathbb{P}) = \int_\mathbb{R} f(x)\, d\mathbb{Q} \tag{27}$$

$$\frac{1}{n}\sum_{i=1}^{n} f(X_i) \xrightarrow{n\to\infty} \int_\mathbb{R} f(x)\, d\mathbb{Q}, \; a.s. \tag{28}$$

# 3 The normal distribution and the central limit theorem

# 4 Swap rates and Valuation of Callable Swaps

$$dX_t = X_t^1 dt + X_t^2 dW_t \tag{29}$$

$$df(X_t) = f'(X_t) dX_t + \frac{1}{2} f''(X_t) d\langle X \rangle_t \tag{30}$$

$$d\langle X \rangle_t = (X_t^2)^2 dt \tag{31}$$

$$L_i(t) = \frac{1}{\delta_i} \frac{P(t, t_{i-1}) - P(t, t_i)}{P(t, t_i)} \tag{32}$$

where $\delta_i \equiv \delta(t_{i-1}, t_i)$ is the coupon length based on the chosen DCC.

$$P_{\text{var}}(t) = N \sum_{i=1}^{n} \mathbb{E}_Q \left( \frac{B_t}{B_{T_i}} \delta_i L_i(\tau_{i-1}) \middle| \mathcal{F}_t \right) \tag{33}$$

$$\begin{aligned} P_{\text{var}}(t) &= N \sum_{i=1}^{n} \delta_i P(t, T_i) \mathbb{E}_{Q_i} \left( L_i(\tau_{i-1}) \middle| \mathcal{F}_t \right) \\ &= N \sum_{i=1}^{n} \delta_i P(t, T_i) L_i(t) \end{aligned} \tag{34}$$

$$P_{\text{fix}}(t) = N \sum_{j=1}^{m} \mathbb{E}_Q \left( \frac{B_t}{B_{T_j}} \delta_j p \middle| \mathcal{F}_t \right) = Np \sum_{j=1}^{m} \delta_j P(t, T_j) \tag{35}$$

$$S^{\pm}(t) = \pm \left( P_{\text{var}}(t) - P_{\text{fix}}(t) \right) \tag{36}$$

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$$p_{\text{fsr}} :\Leftrightarrow S^{\pm} \equiv 0 \tag{37}$$

$$p_{\text{fsr}} = \frac{\sum_{i=1}^{n} \delta_i P(t,T_i) L_i(t)}{\sum_{j=1}^{m} \delta_j P(t,T_j)} \tag{38}$$

Since we can write

$$\sum_{i=1}^{n} \delta_i P(t,T_i) L_i(t) = \sum_{i=1}^{n} \delta_i P(t,T_i) L_i(t) = N \sum_{i=1}^{n} P(t,T_{i-1}) - P(t,T_i) = P(t,T_0) - P(t,T_n) \tag{39}$$

$$p_{\text{fsr}} = \frac{P(t,T_0) - P(t,T_n)}{\sum_{j=1}^{m} \delta_j P(t,T_j)} \tag{40}$$

**Definition 4.1** (Log-normal swap rate model)**.**

$$dp_{\text{fsr}}(t) = p_{\text{fsr}}(t)\sigma(t) dW_t^{1,n} \tag{41}$$

$$B_{\text{swp}}^{\pm}(T_0) := \left(S^{\pm}(T_0)\right)^{+} \tag{42}$$

Using the definition of the foward swap rate $p_{\text{fsr}}$, this can be rewritten as

$$B_{\text{swp}}^{\pm}(T_0) = \left(\pm\left[p_{\text{fsr}}(T_0) - p\right]\right)^{+} N \sum_{j=1}^{m} \delta_j P(T_0,T_j) \tag{43}$$

$$\begin{aligned} B_{\text{swp}}^{\pm}(t) &= \mathbb{E}_Q\left(\frac{B_t}{B_{T_0}}\left(\pm\left[p_{\text{fsr}}(T_0) - p\right]\right)^{+} N \sum_{j=1}^{m} \delta_j P(T_0,T_j) \middle| \mathcal{F}_t\right) \\ &= N \sum_{j=1}^{m} \delta_j P(t,T_j) \mathbb{E}_{Q_{1,n}}\left(\left(\pm\left[p_{\text{fsr}}(T_0) - p\right]\right)^{+} \middle| \mathcal{F}_t\right) \end{aligned} \tag{44}$$

$$\Phi(x) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{1}{2}t^2}\, dt \tag{45}$$

satisfying

$$\Phi(-x) = 1 - \Phi(x) \tag{46}$$

**Theorem 4.2** (Swaption pricing)**.** *In the log-normal model of the forward swap rates, the present value of payer, resp. receiver, swaption is given by*

$$B_{\text{swp}}^{\pm}(t) = N \sum_{j=1}^{m} \delta_j P(t,T_j)\left(p_{\text{fsr}(t)}\Phi(\pm d_1(t)) - p\Phi(\pm d_1(t) \mp \tilde{\sigma})\right) \tag{47}$$

*with*

$$d_1(t) := \frac{\ln\left(\frac{p_{\text{fsr}}(t)}{p}\right) + \frac{1}{2}\tilde{\sigma}^2}{\tilde{\sigma}}, \quad \tilde{\sigma}^2 := \int_t^{T_0} \sigma^2(s)\, ds \tag{48}$$

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**Definition 4.3** (European Callable Swap)**.** A European callable Swap is a Interest rate swap with tenor structure $\{T_i^{\text{fix}}\}_i$ and $\{T_j^{\text{var}}\}_j$ of the fixed and floating leg, respectively, with the possibility, but not the obligation, to quit the swap at a certain time $T_* \in \{T_i^{\text{fix}}\}_i \cap \{T_j^{\text{var}}\}_j$. Hence, at time $t \in [0, T_0]$, the value of the callable swap is given by

$$S^{\pm,\text{call}}(t) = S^{\pm}(t) + B^{\mp}_{\text{swp}}(t) \tag{49}$$

$$\begin{aligned} S^{\pm,\text{call}}(t) = &\pm N \left( P(t,T_0) - P(t,T_*) - p \sum_{j:\, T_j \leq T_*} \delta_j P(t.T_j) \right) \\ &\pm N \sum_{j:\, T_j > T_*} \delta_j P(t,T_j) \left( p_{\text{fsr}(t)} \left[1 \pm \Phi(\mp d_1(t))\right] - p \left[1 \pm \Phi(\mp d_1(t) \pm \tilde{\sigma})\right] \right) \end{aligned} \tag{50}$$

# 5 Feynman-Kac representation and Feynman path integrals

**Theorem 5.1** (The Feynman-Kac representation)**.** *Let $v(t,x) : [0,T] \times \mathbb{R}^n \to \mathbb{R}$ be a solution to the partial differential equation*

$$-\frac{\partial}{\partial t} v(t,x) = (Av)(t,x) - kv(t,x) + g \tag{51}$$

*with operator $A$ defined as*

$$A := \frac{1}{2} a^{ij} \frac{\partial^2}{\partial x^i \partial x^j} + b^i \frac{\partial}{\partial x^i} \tag{52}$$

*with initial condition*

$$v(T,x) = f(x),\ \forall x \in \mathbb{R}^n \tag{53}$$

*Then, $v(t,x)$ can be represented as the expectation value under the risk-free probability measure via*

$$v(t,x) = \mathbb{E}^{t,x} \left( f(X(T)) e^{-\int_t^T k(\tau, X(\tau))\, d\tau} + \int_t^T g(\tau, X(\tau)) e^{-\int_t^\tau k(\tau', X(\tau'))\, d\tau'}\, d\tau \right) \tag{54}$$

**Definition 5.2.** Let $X : I \times \Omega \to \mathcal{E}$ be a stochastic process and

$$\Phi_X : \Omega \to M(I, \mathcal{E}),\ \omega \mapsto X_\bullet(\omega) \tag{55}$$

The Wiener measure $\mu_X$ on the Borel $\sigma$-algebra $\mathcal{B}(M(I,\mathcal{E}))$ is defined as the pushforwad measure

$$\mu_X := \Phi_{X*}\mathbb{P} = \mathbb{P} \circ \Phi_X^{-1} \tag{56}$$

$$I(\{t_i\}_{i=1}^n, \{A_i\}_{i=1}^n) := \left\{ x \in M(I,\mathcal{E}) \,\middle|\, (x(t_i))_{i=1}^n \in \bigtimes_{i=1}^n A_i \right\} \tag{57}$$

**Theorem 5.3.** *Let $W : [0,\infty) \times \Omega \to \mathbb{R}^n$ be a $n$-dimensional Wiener process. Then, on the finite dimensional path sets of $C([0,\infty))$, the associated Wiener measure takes the form*

$$\mu_W(I(\{t_i\}_{i=1}^n, \{A_i\}_{i=1}^n)) = \int_{\bigtimes_{i=1}^n (\alpha_i, \beta_i]} \prod_{i=1}^n \frac{1}{\sqrt{(2\pi)^n (t_i - t_{i-1})}} e^{-\frac{1}{2}\frac{(x_i - x_{i-1})^2}{t_i - t_{i-1}}}\, d^m x_1 \cdots d^m x_n \tag{58}$$

*with $t_0 = 0$ and $x_1 = 0$.*

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By definition of the finite dimensional path sets, one obtains

$$\mu_W(I(\{t_i\}_{i=1}^n, \{A_i\}_{i=1}^n)) = \mathbb{P}\left(\bigcap_{i=1}^n W_{t_i}^{-1}(A_i)\right) \tag{59}$$

By definition of a Wiener process, the increments $W_{t_i} - W_{t_i}$ for $i = 1, \dots, n$ define independent random variables which are normally distributed with mean zero and variance $t_i - t_{i-1}$. Hence, setting

$$\mathbf{W} := \begin{pmatrix} W_{t_1} \\ \vdots \\ W_{t_n} \end{pmatrix} \quad \text{and} \quad \mathbf{Z} := \begin{pmatrix} W_{t_1} \\ W_{t_2} - W_{t_1} \\ \vdots \\ W_{t_n} - W_{t_{n-1}} \end{pmatrix} \tag{60}$$

it follows

$$\mathbf{W} = A^{-1}\mathbf{Z} \tag{61}$$

with

$$A := \begin{pmatrix} 1 & 0 & 0 & & \dots \\ -1 & 1 & 0 & \dots & \\ 0 & -1 & 1 & 0 & \dots \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & & & -1 & 1 \end{pmatrix} \tag{62}$$

Since $\mathbf{Z}$ follows are multivariate normal distribution $\mathbf{Z} \sim \mathcal{N}(0, \Sigma)$ with $\Sigma = \operatorname{diag}(t_1, t_2 - t_1, \dots, t_n - t_{n-1})$, it follows form Lemma ? that $\mathbf{W} \sim \mathcal{N}(0, A^{-1}\Sigma A^{-1T})$. Thus

$$\begin{aligned} \mathbb{P}\left(\bigcap_{i=1}^n W_{t_i}^{-1}(A_i)\right) &= \mathbb{P}\left(\mathbf{W} \in \prod_{i=1}^n A_i\right) = \int_{\prod_{i=1}^n A_i} e^{-\frac{1}{2}(Ax)^T\Sigma^{-1}(Ax)}\, d^m x_1, \dots d^m x_n \\ &= \int_{\prod_{i=1}^n A_i} \prod_{i=1}^n \frac{1}{\sqrt{(2\pi)^n (t_i - t_{i-1})}} e^{-\frac{1}{2}\frac{(x_i - x_{i-1})^2}{t_i - t_{i-1}}}\, d^m x_1 \cdots d^m x_n \end{aligned} \tag{63}$$

# 6 The CEV model

**Definition 6.1** (The CEV model)**.** The (CEV) model for an equity $S_t$ with parameter $\alpha \in [0, 1]$ is given by

$$dS_t = \mu S_t\, dt + \sigma S_t^\alpha\, dW_t. \tag{64}$$

**Definition 6.2** (The $\delta$-dimensional squared Bessel process)**.** Let $\delta \geq 0$ and $x \geq 0$. The unique strong solution $X_t^{(\delta)}$ of the SDE

$$dX_t^{(\delta)} = \delta\, dt + 2\sqrt{|X_t^{(\delta)}|}\, dW_t, \qquad X_0^{(\delta)} = x, \tag{65}$$

is called the $\delta$-dimensional squared Bessel ($\delta$-BESQ) process.

**Theorem 6.3** (CEV as a transform of BESQ)**.** *Let $X^{(\delta)}$ be a $\delta$-dimensional squared Bessel process and define the hitting time*

$$\tau^\delta := \inf\{t \geq 0 : \ X_t^{(\delta)} = 0\}.$$

*Let $v \in \mathbb{R}$ and define the deterministic time change*

$$\tau_t := \frac{\sigma^2}{2v(2-\delta)}\left(1 - e^{-\frac{2v}{2-\delta}t}\right), \qquad (v \neq 0), \tag{66}$$

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*and for $v = 0$ set $\tau_t := \frac{\sigma^2}{(2-\delta)^2} t$. Define*

$$Y_t^{(\delta,v)} := e^{vt} \Big( X^{(\delta)}_{\tau_t \wedge \tau^\delta} \Big)^{1-\frac{\delta}{2}}. \tag{67}$$

*Then $Y^{(\delta,v)}$ is a* weak solution *of a CEV SDE with parameter*

$$\alpha := \frac{1-\delta}{2-\delta}.$$

*Proof.* **Step 1 (Why stopping at $\tau^\delta$).** The map $x \mapsto x^{1-\delta/2}$ is $C^2$ on $(0,\infty)$ but singular at 0 when $\delta > 0$. Hence we apply Itô on $[0, \tau^\delta)$ and freeze at 0 afterwards via $t \wedge \tau^\delta$.

**Step 2 (Itô for the power transform).** Let $\beta := 1 - \delta/2$ and set $Z_u := \big(X^{(\delta)}_{u\wedge\tau^\delta}\big)^\beta$. On $\{u < \tau^\delta\}$ we have $X_u^{(\delta)} > 0$ and Itô's formula applies to $f(x) = x^\beta$:

$$df(X_u) = f'(X_u)\, dX_u + \frac{1}{2} f''(X_u)\, d\langle X\rangle_u.$$

Here $f'(x) = \beta x^{\beta-1}$ and $f''(x) = \beta(\beta-1)x^{\beta-2}$, while $dX_u = \delta\, du + 2\sqrt{X_u}\, dW_u$ and $d\langle X\rangle_u = 4X_u\, du$. Therefore, on $\{u < \tau^\delta\}$,

$$\begin{aligned} dZ_u &= \beta X_u^{\beta-1}\big(\delta\, du + 2\sqrt{X_u}\, dW_u\big) + \frac{1}{2}\beta(\beta-1)X_u^{\beta-2} \cdot 4X_u\, du \\ &= 2\beta X_u^{\beta-\frac{1}{2}}\, dW_u + \beta\delta X_u^{\beta-1}\, du + 2\beta(\beta-1)X_u^{\beta-1}\, du. \end{aligned}$$

Since $\beta = 1 - \delta/2$, we have $\delta + 2(\beta-1) = \delta - \delta = 0$, hence the drift cancels:

$$dZ_u = 2\beta X_u^{\beta-\frac{1}{2}}\, dW_u \qquad (u < \tau^\delta).$$

Using $Z_u = X_u^\beta$ and $\beta - \frac{1}{2} = \frac{1}{2} - \frac{\delta}{2}$ gives

$$X_u^{\beta-\frac{1}{2}} = (X_u^\beta)^{\frac{\beta-\frac{1}{2}}{\beta}} = Z_u^{\frac{\beta-\frac{1}{2}}{\beta}} = Z_u^{\frac{1-\delta}{2-\delta}}.$$

Hence

$$dZ_u = 2\beta\, Z_u^{\frac{1-\delta}{2-\delta}}\, dW_u, \qquad u < \tau^\delta. \tag{68}$$

This is already a CEV-type diffusion (no drift) with exponent $\alpha = \frac{1-\delta}{2-\delta}$.

**Step 3 (Time change and exponential scaling).** Define $Y_t := e^{vt} Z_{\tau_t}$ (with the convention $Z_{\tau_t} := Z_{\tau_t \wedge \tau^\delta}$). Since $\tau_t$ is deterministic and increasing, we can write for a Brownian motion $\widetilde{W}$ (by standard time-change arguments)

$$dZ_{\tau_t} = 2\beta\, Z_{\tau_t}^\alpha \sqrt{\tau_t'}\, d\widetilde{W}_t, \qquad \alpha = \frac{1-\delta}{2-\delta},$$

where $\tau_t' = \frac{d}{dt}\tau_t$. Using the product rule for $Y_t = e^{vt} Z_{\tau_t}$:

$$dY_t = vY_t\, dt + e^{vt}\, dZ_{\tau_t} = vY_t\, dt + 2\beta e^{vt} Z_{\tau_t}^\alpha \sqrt{\tau_t'}\, d\widetilde{W}_t.$$

But $Z_{\tau_t} = e^{-vt} Y_t$, hence $e^{vt} Z_{\tau_t}^\alpha = e^{vt} e^{-\alpha vt} Y_t^\alpha = e^{(1-\alpha)vt} Y_t^\alpha$. Choose $\tau_t$ such that

$$2\beta \sqrt{\tau_t'}\, e^{(1-\alpha)vt} = \sigma,$$

which is equivalent to

$$\tau_t' = \frac{\sigma^2}{4\beta^2}\, e^{-2(1-\alpha)vt}.$$

Using $\beta = 1 - \delta/2$ and $1 - \alpha = \frac{1}{2-\delta}$, this integrates to (70). Therefore

$$dY_t = vY_t\, dt + \sigma Y_t^\alpha\, d\widetilde{W}_t,$$

i.e. $Y$ is a weak solution of a CEV SDE with exponent $\alpha = \frac{1-\delta}{2-\delta}$. □

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# 7 The CIR model

**Definition 7.1** (The CIR model)**.** The Cox-Ingersoll-Ross (CIR) model for the short rate $r_t$ is

$$dr_t = \kappa(\theta - r_t)\,dt + \sigma\sqrt{r_t}\,dW_t, \qquad r_0 \geq 0, \tag{69}$$

with mean reversion speed $\kappa > 0$, long-term mean $\theta > 0$, and volatility $\sigma > 0$.

**Lemma 7.2.** *Let $X^{(\delta)}$ be a $\delta$-dimensional squared Bessel process with $\delta := \frac{4\kappa\theta}{\sigma^2}$. Define the hitting time*

$$\tau^\delta := \inf\{t \geq 0 : \; X_t^{(\delta)} = 0\}.$$

*Let $v \in \mathbb{R}$ and define the deterministic time change*

$$\tau_t := \frac{\sigma^2}{4\kappa}\left(e^{\kappa t} - 1\right) \tag{70}$$

*Then the stochastic process $Y_t$ defined via*

$$Y_t := e^{-\kappa t} X^{(\delta)}_{\tau_t \wedge \tau^\delta} \tag{71}$$

*is a* weak solution *of the CIR SDE.*

*Proof.*

$$\begin{aligned}
dY_t &= de^{-\kappa t} X^{(\delta)}_{\tau_t\wedge\tau^\delta} + e^{-\kappa t} dX^{(\delta)}_{\tau_t\wedge\tau^\delta} \\
&= -\kappa Y_t dt + e^{-\kappa t}\left(\frac{4\kappa\theta}{\sigma^2}d\tau + 2\sqrt{|X^{(\delta)}_{\tau_t\wedge\tau^\delta}|}dW_{\tau_t\wedge\tau^\delta}\right) \\
&= \kappa(\theta - Y_t)dt + 2e^{-\frac{\kappa}{2}t}\sqrt{Y}_t dW_{\tau_t\wedge\tau^\delta} \\
&= \kappa(\theta - Y_t)dt + \sigma dW_t^{(\delta)}
\end{aligned} \tag{72}$$

where we defined

$$W_t^{(\delta)} = \int_0^{\tau_t} \frac{2}{\sigma\sqrt{1 + \frac{4\kappa u}{\sigma^2}}} dW_u \tag{73}$$

The theorem follows if we can show that $W_t^{(\delta)}$ defines a Wiener process. By definition, $W_t^{(\delta)}$ is a continuous Martingale. Moreover, for the quadratic variation, we find

$$< W^{(\delta)} >_t = \frac{4}{\sigma^2}\int_0^{\tau_t} \frac{1}{1 + \frac{4\kappa u}{\sigma^2}} du = \frac{1}{\kappa}\ln\left(1 + \frac{4\kappa\tau_t}{\sigma^2}\right) = t \tag{74}$$

□

**Theorem 7.3** (CIR transition law)**.** *Let $r_t$ be a solution to the the CIR SDE. For each $t \geq 0$ and $\Delta > 0$, conditionally on $r_t = r$,*

- *$r_{t+\Delta}$ is distributed according to a scaled non-central $\chi^2$ distribution:*

$$r_{t+\Delta} \sim c\,\chi^2_d(\lambda), \tag{75}$$

*where*

$$c = \frac{\sigma^2}{4\kappa}(1 - e^{-\kappa\Delta}), \qquad d = \frac{4\kappa\theta}{\sigma^2}, \qquad \lambda = \frac{4\kappa}{\sigma^2(e^{\kappa\Delta} - 1)}\,r.$$

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- *In particular, the conditional expectation and variance are given by*

$$\mathbb{E}[r_{t+\Delta} \mid r_t = r] = \theta + (r-\theta)e^{-\kappa\Delta}, \tag{76}$$

$$\mathrm{Var}(r_{t+\Delta} \mid r_t = r) = r\,\frac{\sigma^2}{\kappa}e^{-\kappa\Delta}(1-e^{-\kappa\Delta}) + \theta\,\frac{\sigma^2}{2\kappa}(1-e^{-\kappa\Delta})^2. \tag{77}$$

*Proof.* According to lemma ??, we know that $r_t$ can be written in the form

$$r_t = e^{-\kappa t}X^{(\delta)}_{\tau_t} \tag{78}$$

with $X^{(\delta)}_{\tau_t}$ a $\delta$-dimensional quadratic Bessel process with $\delta = \frac{4\kappa\theta}{\sigma^2}$. According to theorem ??, we know that $X^{(\delta)}_{\tau_t}$ with the initial condition $X^{(\delta)}_0 = r_0$ is distributed according to a scaled non-central $\chi^2$ distribution, i.e.,

$$X_{\tau_t} \sim \tau_t\,\chi^2_\delta\left(\frac{r}{\tau_t}\right),$$

Hence, conditional on $r_t$, the distribution of the time shifted process $r_{t+\Delta}$ is given by

$$r_{t+\Delta}|r_t = e^{-\kappa(t+\Delta)}X^{(\delta)}_{\tau_{t+\Delta}}|X^{(\delta)}_{\tau_t} \sim e^{-\kappa(t+\Delta)}\Delta\tau_t\chi^2_\delta\left(e^{\kappa t}\frac{r_t}{\Delta\tau_t}\right) = \frac{\sigma^2}{4\kappa}(1-e^{-\kappa\Delta})\chi^2_\delta\left(\frac{4\kappa}{\sigma^2}\frac{r_t}{e^{\kappa\Delta}-1}\right).$$

with $\Delta\tau_t = \tau_{t+\Delta} - \tau_t = \frac{\sigma^2}{4\kappa}e^{\kappa t}(e^{\kappa\Delta}-1)$. Using the known moments of $\chi^2_\delta(\lambda)$ (see ??) with $\lambda = \frac{4\kappa}{\sigma^2}\frac{r_t}{e^{\kappa\Delta}-1}$ together with the scaling $c := \frac{\sigma^2}{4\kappa}(1-e^{-\kappa\Delta})$ gives

$$\mathbb{E}[r_{t+\Delta} \mid r_t = r] = c(d+\lambda), \qquad \mathrm{Var}(r_{t+\Delta} \mid r_t = r) = c^2\,2(d+2\lambda),$$

which simplifies to (76)–(77). □

**Theorem 7.4** (Quadratic Exponential Scheme for $V$ (Andersen 2007))**.** *Fix a time step $\Delta > 0$ and condition on $V_t = v \geq 0$. Let*

$$m := \mathbb{E}[V_{t+\Delta} \mid V_t = v] = \theta + (v-\theta)e^{-\kappa\Delta}, \tag{79}$$

$$s^2 := \mathrm{Var}(V_{t+\Delta} \mid V_t = v) = v\,\frac{\varepsilon^2}{\kappa}e^{-\kappa\Delta}(1-e^{-\kappa\Delta}) + \theta\,\frac{\varepsilon^2}{2\kappa}(1-e^{-\kappa\Delta})^2, \tag{80}$$

$$\psi := \frac{s^2}{m^2}. \tag{81}$$

*Choose a threshold $\psi_c \in [1,2]$ (Andersen recommends $\psi_c = 1.5$).*

***Case 1 (Quadratic-Gaussian approximation).*** *If $\psi \leq \psi_c$, define*

$$b^2 := \frac{2}{\psi} - 1 + \sqrt{\frac{2}{\psi}}\sqrt{\frac{2}{\psi}-1}, \tag{82}$$

$$a := \frac{m}{1+b^2}. \tag{83}$$

*If $Z \sim N(0,1)$, set*

$$\widehat{V}_{t+\Delta} := a\,(b+Z)^2. \tag{84}$$

***Case 2 (Point-mass + exponential tail).*** *If $\psi > \psi_c$, define*

$$p := \frac{\psi-1}{\psi+1} \in [0,1), \tag{85}$$

$$\beta := \frac{1-p}{m} = \frac{2}{m(\psi+1)} > 0. \tag{86}$$

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*If* $U \sim \mathrm{Unif}(0,1)$*, set*

$$\widehat{V}_{t+\Delta} := \begin{cases} 0, & U \le p, \\ -\frac{1}{\beta}\ln\left(\frac{1-U}{1-p}\right), & U > p. \end{cases} \tag{87}$$

*Then in both cases the approximation* $\widehat{V}_{t+\Delta}$ *matches the first two conditional moments:*

$$\mathbb{E}[\widehat{V}_{t+\Delta} \mid V_t = v] = m, \qquad \mathrm{Var}(\widehat{V}_{t+\Delta} \mid V_t = v) = s^2.$$

*Proof.* We prove the stated moment-matching claims for the two branches.

**Step 1: Conditional moments** $m$ **and** $s^2$**.** The process $V$ is CIR, hence its conditional mean and variance are given by (79)–(80) (cf. Theorem 7.3 in the CIR section with $r \mapsto V$, $\sigma \mapsto \varepsilon$). Define $\psi = s^2/m^2$ as in (81).

**Step 2: Case 1 (quadratic form).** Assume $\psi \le \psi_c \le 2$ and define $\widehat{V} = a(b+Z)^2$ with $Z \sim N(0,1)$. Then $(b+Z)^2$ has a non-central $\chi^2$ distribution with one degree of freedom and non-centrality parameter $b^2$. Therefore,

$$\mathbb{E}[(b+Z)^2] = 1 + b^2, \qquad \mathrm{Var}((b+Z)^2) = 2(1+2b^2).$$

Hence

$$\mathbb{E}[\widehat{V}] = a(1+b^2), \qquad \mathrm{Var}(\widehat{V}) = 2a^2(1+2b^2).$$

The choice $a = m/(1+b^2)$ in (83) enforces $\mathbb{E}[\widehat{V}] = m$. With this $a$, the variance condition $\mathrm{Var}(\widehat{V}) = s^2$ is equivalent to

$$2\frac{m^2}{(1+b^2)^2}(1+2b^2) = s^2 \quad \Longleftrightarrow \quad \psi = \frac{s^2}{m^2} = 2\frac{1+2b^2}{(1+b^2)^2}.$$

Solving this quadratic equation for $b^2$ yields exactly (82); a real solution exists whenever $\psi \le 2$. Therefore Case 1 matches $m$ and $s^2$.

**Step 3: Case 2 (mass at zero + exponential tail).** Assume $\psi > \psi_c \ge 1$ and define $\widehat{V}$ by (87). This corresponds to the mixture distribution

$$\mathbb{P}(\widehat{V} = 0) = p, \qquad \widehat{V} \mid \{\widehat{V} > 0\} \sim \mathrm{Exp}(\beta),$$

with density $(1-p)\beta e^{-\beta x}$ on $x > 0$. Its moments are

$$\mathbb{E}[\widehat{V}] = (1-p)\frac{1}{\beta}, \qquad \mathbb{E}[\widehat{V}^2] = (1-p)\frac{2}{\beta^2}, \qquad \mathrm{Var}(\widehat{V}) = \frac{(1-p)(1+p)}{\beta^2} = \frac{1-p^2}{\beta^2}.$$

Imposing $\mathbb{E}[\widehat{V}] = m$ and $\mathrm{Var}(\widehat{V}) = s^2$ gives the system

$$\frac{1-p}{\beta} = m, \qquad \frac{1-p^2}{\beta^2} = s^2.$$

Divide the second equation by the square of the first:

$$\frac{1-p^2}{(1-p)^2} = \frac{s^2}{m^2} = \psi \quad \Longleftrightarrow \quad \frac{1+p}{1-p} = \psi \quad \Longleftrightarrow \quad p = \frac{\psi-1}{\psi+1},$$

which is (85) and satisfies $p \in [0,1)$ iff $\psi \ge 1$. Then $\beta = (1-p)/m$ gives (86). Thus Case 2 also matches $m$ and $s^2$.

Combining both cases proves the theorem. □

**Remark 7.5** (Fuzzy switching for differentiability (implementation note))**.** In Monte Carlo with AAD, the hard branch $\mathbf{1}_{\{\psi \le \psi_c\}}$ can create non-differentiabilities. A practical remedy is to replace it by a smooth weight $w(\psi) \in [0,1]$ (e.g. linear or sigmoid smoothing around $\psi_c$) and set $\widehat{V} = (1-w)\widehat{V}^{(1)} + w\,\widehat{V}^{(2)}$. This keeps the scheme numerically stable and yields smooth pathwise sensitivities.

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# 8 The Heston Model

**Definition 8.1** (The Heston Model)**.** Let $(W_t^{(1)}, W_t^{(2)})$ be a two-dimensional Brownian motion with correlation $\langle W^{(1)}, W^{(2)}\rangle_t = \rho t$, $\rho \in [-1, 1]$. The *Heston stochastic volatility model* under the risk-neutral measure $\mathbb{Q}$ (with zero dividend yield for simplicity) is given by

$$dS_t = rS_t\,dt + \sqrt{V_t}\,S_t\,dW_t^{(1)}, \tag{88}$$

$$dV_t = \kappa(\theta - V_t)\,dt + \varepsilon\sqrt{V_t}\,dW_t^{(2)}, \tag{89}$$

with parameters $r \in \mathbb{R}$, $\kappa > 0$, $\theta > 0$, $\varepsilon > 0$, $V_0 \geq 0$, and $S_0 > 0$.

**Remark 8.2** (CIR variance process)**.** The variance process $V_t$ in (89) is a CIR process. In particular, $V_t \geq 0$ a.s. If $2\kappa\theta \geq \varepsilon^2$ (Feller condition), then $V_t > 0$ a.s. for all $t > 0$.

**Theorem 8.3** (QE discretization for $\log S$ (Andersen 2007))**.** *Consider the Heston model in Definition 8.1 and fix a step $\Delta > 0$. Assume that $V_t$ is given and that an approximation $\widehat{V}_{t+\Delta}$ has been generated (e.g. by Theorem 7.4). Define the integrated variance approximation*

$$\int_t^{t+\Delta} V_u\,du \;\approx\; \Delta\big(\gamma_1 V_t + \gamma_2 \widehat{V}_{t+\Delta}\big), \tag{90}$$

*for constants $\gamma_1, \gamma_2 \in [0, 1]$ (typical choices: Euler $\gamma_1 = 1, \gamma_2 = 0$, or trapezoidal $\gamma_1 = \gamma_2 = \frac{1}{2}$). Then a consistent discretization of $\log S$ over $[t, t + \Delta]$ is*

$$\log \widehat{S}_{t+\Delta} = \log S_t + K_0 + K_1 V_t + K_2 \widehat{V}_{t+\Delta} + \sqrt{K_3 V_t + K_4 \widehat{V}_{t+\Delta}}\;Z, \tag{91}$$

*where $Z \sim N(0, 1)$ is independent of all random numbers used to generate $\widehat{V}_{t+\Delta}$ and*

$$K_0 = r\Delta - \frac{\rho\kappa\theta}{\varepsilon}\Delta, \tag{92}$$

$$K_1 = \gamma_1\Delta\Big(\frac{\kappa\rho}{\varepsilon} - \frac{1}{2}\Big) - \frac{\rho}{\varepsilon}, \tag{93}$$

$$K_2 = \gamma_2\Delta\Big(\frac{\kappa\rho}{\varepsilon} - \frac{1}{2}\Big) + \frac{\rho}{\varepsilon}, \tag{94}$$

$$K_3 = \gamma_1\Delta(1 - \rho^2), \tag{95}$$

$$K_4 = \gamma_2\Delta(1 - \rho^2). \tag{96}$$

*Proof.* Write the Heston dynamics in log form. By Itô's formula applied to $\log S_t$,

$$d\log S_t = \Big(r - \tfrac{1}{2}V_t\Big)dt + \sqrt{V_t}\,dW_t^{(1)}. \tag{97}$$

Use the standard decomposition of correlated Brownian motions:

$$dW_t^{(1)} = \rho\,dW_t^{(2)} + \sqrt{1 - \rho^2}\,dW_t^{\perp},$$

where $W^{\perp}$ is a Brownian motion independent of $W^{(2)}$. Then

$$d\log S_t = \Big(r - \tfrac{1}{2}V_t\Big)dt + \rho\sqrt{V_t}\,dW_t^{(2)} + \sqrt{1 - \rho^2}\sqrt{V_t}\,dW_t^{\perp}. \tag{98}$$

Next, use the variance SDE $dV_t = \kappa(\theta - V_t)dt + \varepsilon\sqrt{V_t}\,dW_t^{(2)}$ to express $\sqrt{V_t}\,dW_t^{(2)}$ in terms of $dV_t$:

$$\sqrt{V_t}\,dW_t^{(2)} = \frac{1}{\varepsilon}\,dV_t - \frac{\kappa(\theta - V_t)}{\varepsilon}\,dt. \tag{99}$$

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Insert (99) into (98) to obtain

$$\begin{aligned} d\log S_t &= \Big(r - \tfrac{1}{2}V_t\Big)dt + \frac{\rho}{\varepsilon}\,dV_t - \frac{\rho\kappa(\theta - V_t)}{\varepsilon}\,dt + \sqrt{1-\rho^2}\sqrt{V_t}\,dW_t^{\perp} \\ &= \Big(r - \frac{\rho\kappa\theta}{\varepsilon}\Big)dt + \Big(\frac{\rho\kappa}{\varepsilon} - \frac{1}{2}\Big)V_t\,dt + \frac{\rho}{\varepsilon}\,dV_t + \sqrt{1-\rho^2}\sqrt{V_t}\,dW_t^{\perp}. \end{aligned} \tag{100}$$

Integrate (100) from $t$ to $t+\Delta$:

$$\log S_{t+\Delta} - \log S_t = \Big(r - \frac{\rho\kappa\theta}{\varepsilon}\Big)\Delta + \Big(\frac{\rho\kappa}{\varepsilon} - \frac{1}{2}\Big)\int_t^{t+\Delta} V_u\,du + \frac{\rho}{\varepsilon}(V_{t+\Delta} - V_t) + \sqrt{1-\rho^2}\int_t^{t+\Delta}\sqrt{V_u}\,dW_u^{\perp}. \tag{101}$$

Approximate the time integral by (90) and replace $V_{t+\Delta}$ by $\widehat{V}_{t+\Delta}$ (from the variance step). Conditional on $V_t$ and $\widehat{V}_{t+\Delta}$, the Itô integral in (101) is Gaussian with mean 0 and variance $\int_t^{t+\Delta} V_u\,du$, which we approximate by $\Delta(\gamma_1 V_t + \gamma_2\widehat{V}_{t+\Delta})$. Hence we set

$$\sqrt{1-\rho^2}\int_t^{t+\Delta}\sqrt{V_u}\,dW_u^{\perp} \approx \sqrt{(1-\rho^2)\Delta(\gamma_1 V_t + \gamma_2\widehat{V}_{t+\Delta})}\;Z, \quad Z \sim N(0,1).$$

Collecting coefficients of $\Delta$, $V_t$, $\widehat{V}_{t+\Delta}$ and the Gaussian term yields exactly (91) with (92)–(96). □

# 9 Credit Derivatives and Counterparty Credit Risk

## 9.1 Reduced form models

**Definition 9.1** (Poisson process)**.**

**Theorem 9.2** (Change of filtration)**.** *Let $\mathcal{G}_t := \mathcal{F}_t \vee \sigma(\{\tau \le u\}, u \le t)$ be the progressive enlargement of $\mathbb{F} = (\mathcal{F}_t)_{t\ge 0}$ with the default time $\tau$. Define*

$$G_t := \mathbb{Q}(\tau > t \mid \mathcal{F}_t), \qquad t \ge 0,$$

*and assume $G_t > 0$ a.s. for all $t$. Then for any $t \le T$ and any integrable $\mathcal{F}_T$-measurable random variable $X$,*

$$\mathbb{E}\big(\mathbb{1}_{\{\tau>T\}}X \mid \mathcal{G}_t\big) = \frac{\mathbb{1}_{\{\tau>t\}}}{\mathbb{Q}(\tau > t \mid \mathcal{F}_t)}\,\mathbb{E}\big(\mathbb{1}_{\{\tau>T\}}X \mid \mathcal{F}_t\big)\,. \tag{102}$$

*Proof.* Fix $t \le T$ and set $G_t = \mathbb{Q}(\tau > t \mid \mathcal{F}_t)$. We will show that the random variable

$$Y_t := \frac{\mathbb{1}_{\{\tau>t\}}}{G_t}\,\mathbb{E}\big(\mathbb{1}_{\{\tau>T\}}X \mid \mathcal{F}_t\big)$$

satisfies the defining property of the conditional expectation $\mathbb{E}(\mathbb{1}_{\{\tau>T\}}X \mid \mathcal{G}_t)$.

Let $A \in \mathcal{G}_t$. Since $\mathcal{G}_t = \mathcal{F}_t \vee \sigma(\tau\wedge t)$ and $\sigma(\tau\wedge t)$ is generated by the events $\{\tau > t\}$ and $\{\tau \le t\}$ together with $\mathcal{F}_t$, it suffices (by a standard $\pi$–$\lambda$ argument) to verify the identity for sets of the form

$$A = B \cap \{\tau > t\} \quad \text{and} \quad A = B \cap \{\tau \le t\}, \qquad B \in \mathcal{F}_t.$$

**Case 1:** $A = B \cap \{\tau \le t\}$**.** On $\{\tau \le t\}$ we have $\mathbb{1}_{\{\tau>T\}} = 0$ (because $t \le T$), hence

$$\mathbb{E}\big(\mathbb{1}_A\,\mathbb{1}_{\{\tau>T\}}X\big) = 0.$$

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On the other hand, $\mathbb{1}_{\{\tau>t\}} = 0$ on $A$, so $\mathbb{1}_A Y_t = 0$ and therefore

$$\mathbb{E}(\mathbb{1}_A Y_t) = 0.$$

Thus the desired identity holds in this case.

**Case 2:** $A = B \cap \{\tau > t\}$**.** Since $\{\tau > T\} \subseteq \{\tau > t\}$ for $t \leq T$, we have

$$\mathbb{1}_A \mathbb{1}_{\{\tau>T\}} = \mathbb{1}_B \mathbb{1}_{\{\tau>T\}}.$$

Hence, using the tower property for $\mathcal{F}_t \subseteq \mathcal{F}_T$,

$$\begin{aligned}\mathbb{E}\big(\mathbb{1}_A\, \mathbb{1}_{\{\tau>T\}} X\big) &= \mathbb{E}\big(\mathbb{1}_B\, \mathbb{1}_{\{\tau>T\}} X\big) \\ &= \mathbb{E}\big(\mathbb{1}_B\, \mathbb{E}\big(\mathbb{1}_{\{\tau>T\}} X \mid \mathcal{F}_t\big)\big)\,.\end{aligned}$$

Now compute $\mathbb{E}(\mathbb{1}_A Y_t)$:

$$\mathbb{E}(\mathbb{1}_A Y_t) = \mathbb{E}\bigg(\mathbb{1}_B \mathbb{1}_{\{\tau>t\}} \frac{1}{G_t}\, \mathbb{E}\big(\mathbb{1}_{\{\tau>T\}} X \mid \mathcal{F}_t\big)\bigg)\,.$$

Condition on $\mathcal{F}_t$ and use that $\mathbb{1}_B$, $G_t^{-1}$, and $\mathbb{E}(\mathbb{1}_{\{\tau>T\}} X \mid \mathcal{F}_t)$ are $\mathcal{F}_t$-measurable:

$$\begin{aligned}\mathbb{E}(\mathbb{1}_A Y_t) &= \mathbb{E}\bigg(\mathbb{1}_B \frac{1}{G_t}\, \mathbb{E}\big(\mathbb{1}_{\{\tau>T\}} X \mid \mathcal{F}_t\big)\, \mathbb{E}(\mathbb{1}_{\{\tau>t\}} \mid \mathcal{F}_t)\bigg) \\ &= \mathbb{E}\bigg(\mathbb{1}_B \frac{1}{G_t}\, \mathbb{E}\big(\mathbb{1}_{\{\tau>T\}} X \mid \mathcal{F}_t\big)\, G_t\bigg) \\ &= \mathbb{E}\big(\mathbb{1}_B\, \mathbb{E}\big(\mathbb{1}_{\{\tau>T\}} X \mid \mathcal{F}_t\big)\big)\,.\end{aligned}$$

Comparing with the expression for $\mathbb{E}(\mathbb{1}_A \mathbb{1}_{\{\tau>T\}} X)$ above yields

$$\mathbb{E}(\mathbb{1}_A Y_t) = \mathbb{E}\big(\mathbb{1}_A \mathbb{1}_{\{\tau>T\}} X\big) \quad \text{for all } A \in \mathcal{G}_t.$$

Therefore $Y_t = \mathbb{E}(\mathbb{1}_{\{\tau>T\}} X \mid \mathcal{G}_t)$ a.s., which is exactly (102). □

$$\tau := \inf\{t \geq 0 | \Lambda(t) \geq E\} \tag{103}$$

with $E \sim \text{Exp}(1)$. Hence, by definition, it follows

$$\{\Lambda(\tau) > \Lambda(t)\} = \{E > \Lambda(t)\} \tag{104}$$

By the substition lemma, we have

$$\begin{aligned}\mathbb{Q}(\tau > t | \mathcal{F}_t) &= \mathbb{E}(\mathbb{1}_{\{\Lambda(\tau)>\Lambda(t)\}} | \mathcal{F}_t) \\ &= \mathbb{E}(\mathbb{1}_{\{E>\Lambda(t)\}} | \mathcal{F}_t) \\ &= e^{-\Lambda(t)}\end{aligned} \tag{105}$$

Using the tower property, we thus find

$$\mathbb{Q}(\tau > t) = \mathbb{E}(\mathbb{E}(\mathbb{1}_{\{E>\Lambda(t)\}} | \mathcal{F}_t)) = \mathbb{E}(e^{-\Lambda(t)}) \tag{106}$$

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## 9.2 Credit Default Swaps valuation

**Definition 9.3** (Credit Default Swap with running payments)**.** A *credit default swap with running payments (RCDS)* is a bilateral financial contract between two counterparties: the protection buyer $A$ and the protection seller $B$, written on an underlying reference entity $C$.

Let $[T_a, T_b]$ be the contractual protection period and let $\tau$ denote the default time of $C$. If a credit event (default) of $C$ occurs at time $\tau \in (T_a, T_b]$, the protection seller $B$ pays the protection buyer $A$ a protection payment equal to the *loss given default (LGD)*,

$$\mathrm{LGD} = 1 - R,$$

with $R$ the recovery rate.

In return, the protection buyer $A$ pays the protection seller $B$ a running premium at a fixed rate $S$ (the CDS spread) of the form

$$(T_i - T_{i-1})S \tag{107}$$

on a set of predetermined payment dates $T_0 = T_a < T_1 < \cdots < T_n = T_b$ Premium payments are made up to the earlier of default or maturity and include accrued premium

$$(t - T_{\beta(t)-1})S \tag{108}$$

from the last payment date $T_{\beta(t)-1}$ up to the default time $\tau$ if default occurs between payment dates, i.e., $\beta(t) := \min\{i | T_i \geq t\}$.

From the definition, it follows that the payoff of a RCDS as regarded from the protection seller is given by

$$\begin{aligned}\Pi_{\mathrm{RCDS}} = &\sum_{i=1}^{n} D(t, T_i)S(T_i - T_i)\mathbb{1}_{\{t \geq T_i\}} + D(t,\tau)S(\tau - T_{\beta(\tau)-1})\mathbb{1}_{\{T_a < \tau \leq T_b\}} \\ &- (1-R)D(t,\tau)\mathbb{1}_{\{t<T\}}\end{aligned} \tag{109}$$

## 9.3 Collateralized Debt Obligations – The face of the GFC

### 9.3.1 Basic concepts and motivation

Collateralized Debt Obligations (CDOs) became one of the most prominent credit derivatives during the Global Financial Crisis. The key economic idea is that a bank can transfer (parts of) its credit risk to investors by issuing securities whose cashflows depend on the performance of an underlying credit portfolio. This allows the bank to reduce retained risk (and hence regulatory capital) while investors are compensated via a spread/premium for bearing default risk.

**From single-name risk transfer to tranching.** A natural step to make credit risk transfer scalable is to *package* a portfolio of credits (or CDS) and to *tranche* the resulting portfolio losses into several layers (tranches). Losses are allocated sequentially: first to the most junior (equity) tranche, then to mezzanine tranches, and finally to senior tranches. This introduces different *seniorities* and therefore different risk/return profiles.

**Cash vs. synthetic CDOs.** If the structure is backed by *physical* loans/bonds ("names"), one speaks of a *cash CDO*. In contrast, a *synthetic CDO* references a portfolio of CDS (often on a standardized index), where the protection buyer/seller exchange default-contingent payments against periodic premiums.

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### 9.3.2 Tranche loss as a function of portfolio loss

**Definition 9.4** (Portfolio loss process)**.** Consider a portfolio of $N$ names with default times $\tau_i$ and (fractional) loss-given-default $\mathrm{LGD}_i \in (0,1]$ (often $\mathrm{LGD}_i = 1 - \mathrm{Rec}_i$). The (fractional) portfolio loss up to time $t$ is defined as

$$\mathrm{Loss}(t) := \frac{1}{N} \sum_{i=1}^{N} \mathrm{LGD}_i \, \mathbb{1}_{\{\tau_i \le t\}}. \tag{110}$$

In the homogeneous case $\mathrm{LGD}_i \equiv \mathrm{LGD}$ this reduces to $\mathrm{Loss}(t) = \mathrm{LGD} \cdot \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}_{\{\tau_i \le t\}}$.

**Definition 9.5** (Tranche loss for attachment/detachment points)**.** Let $0 \le A < B \le 1$ denote the *attachment* and *detachment* point. The *tranched loss* (normalized to the tranche thickness $B - A$) is

$$\begin{aligned} L^{A,B}(t) :=& \frac{1}{B-A} \left( (\mathrm{Loss}(t) - A) \mathbb{1}_{\{A \le \mathrm{Loss}(t) < B\}} + (B - A) \mathbb{1}_{\{\mathrm{Loss}(t) \ge B\}} \right) \\ =& \frac{(\mathrm{Loss}(t) - A)^+ - (\mathrm{Loss}(t) - B)^+}{B - A} \end{aligned} \tag{111}$$

Equivalently, using $L^{0,x}(t) := \min\{\mathrm{Loss}(t)/x, 1\}$,

$$L^{A,B}(t) = \frac{1}{B-A} \left( B L^{0,B}(t) - A L^{0,A}(t) \right). \tag{112}$$

**Interpretation.** If $\mathrm{Loss}(t) < A$, the tranche has no loss. If $A \le \mathrm{Loss}(t) < B$, the tranche loss equals $\frac{\mathrm{Loss}(t) - A}{B - A}$. If $\mathrm{Loss}(t) \ge B$, the tranche is fully exhausted ($L^{A,B}(t) = 1$).

### 9.3.3 Protection and premium legs (cashflow view)

A synthetic CDO tranche can be seen as a CDS written on the tranche loss.

**Protection leg.** The protection seller pays increments of tranche loss whenever defaults cause $\mathrm{Loss}(t)$ to jump upward. Let $\{\tau^{(1)}, \dots, \tau^{(n)}\}$ be the ordered default times in the portfolio (order statistics). The discrete protection payment at the $k$-th default time $\tau^{(k)}$ can be written as

$$\Delta L^{A,B}\left(\tau^{(k)}\right) = L^{A,B}\left(\tau^{(k)}\right) - L^{A,B}\left(\tau^{(k-1)}\right), \qquad \tau^{(0)} := t. \tag{113}$$

Hence, the payoff of the protection leg can be written as

$$\Pi_{\mathrm{CDO}}^{\mathrm{prot}} = \sum_{i=1}^{n} D(0, \tau_i) \Delta L^{A,B}(\tau_i) \tag{114}$$

Generally, given any equidistant partition $t_i := i\frac{T-t}{n}$, $i = 0, \dots, n$, $n \in \mathbb{N}$, of the interval $[t, T]$, let us set $\Delta L^{A,B}(t_k) := L^{A,B}(t_k) - L^{A,B}(t_{k-1})$ for $k = 1, \dots, n$. Then, since the tranche loss only jumps at default times, it follows that the payoff of the protection leg can be written as the limit

$$\Pi_{\mathrm{CDO}}^{\mathrm{prot}} = \lim_{n \to \infty} \sum_{k=1}^{n} \delta^{(n)} D(0, t_k) \Delta L^{A,B}(t_k) =: \int_0^T D(0,t) dL^{A,B}(t) \tag{115}$$

with $\delta^{(n)} := \frac{T-t}{n}$.

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**Premium leg and outstanding tranche notional.** Let $R$ denote the tranche premium rate (spread). The premium is paid periodically on the *outstanding* tranche notional, which decreases as the tranche incurs losses. In normalized units, the outstanding notional is

$$\mathrm{Outst}^{A,B}(t) := 1 - L^{A,B}(t). \tag{116}$$

Thus, premiums accrue on the *survived* (not yet lost) tranche notional, not on the full original notional. The payoff of the remium leg therefore reads

$$\Pi_{\mathrm{CDO}}^{\mathrm{premium}} = \sum_{i=1}^{N} D(0,T_i)\delta_i S^{A,B}\mathrm{Outst}^{A,B}(T_i) = S^{A,B}\sum_{i=1}^{N} D(0,T_i)\delta_i \left(1 - L^{A,B}(T_i)\right) \tag{117}$$

Hence, in total, the present value of the synthetic CDO is given by

$$\mathrm{PV}_{\mathrm{CDO}} := \mathbb{E}\left(\Pi_{\mathrm{CDO}}^{\mathrm{prot}} - \Pi_{\mathrm{CDO}}^{\mathrm{premium}} | \mathcal{F}_t\right) \tag{118}$$

The fair spread $\mathrm{PV}_{\mathrm{CDO}}$ of the CDO is defined as the tranche premium rate that evaluates the present value to zero at the start of the contract. Hence, it follows

$$S_t^{A,B} = \frac{\int_t^T D(0,s)d\mathbb{E}(L^{A,B}(s)|\mathcal{F}_t)}{\sum_{i=1}^{N} D(0,T_i)\delta_i \left(1 - \mathbb{E}(L^{A,B}(T_i)|\mathcal{F}_t)\right)} \tag{119}$$

### 9.3.4 Standardized index tranches (iTraxx / CDX)

In practice, standardized tranche quotes exist for major CDS indices such as iTraxx (Europe) and CDX (North America). The iTraxx index references a standardized basket of 125 equally weighted single-name CDS on European investment-grade corporates, each with identical maturity, contractual terms, and loss-given-default, making the index (and its tranches) particularly amenable to homogeneous portfolio credit modelling. For iTraxx, the attachment points (percentages of portfolio notional) are

$$A_1 = 0, \quad A_2 = 3\%, \quad A_3 = 6\%, \quad A_4 = 9\%, \quad A_5 = 12\%, \quad A_6 = 22\%. \tag{120}$$

This yields tranches $(A_1, A_2)$, $(A_2, A_3)$, $\ldots$, $(A_5, A_6)$ (and often a senior tranche above $A_6$ depending on market convention). For the CDX index, typical standardized attachment points are

$$A_1 = 0, \quad A_2 = 3\%, \quad A_3 = 7\%, \quad A_4 = 10\%, \quad A_5 = 15\%, \quad A_6 = 30\%. \tag{121}$$

**Example 9.6** (iTraxx merzanine tranche (3%–6%).)**.** Consider selling protection on the $3\%-6\%$ tranche with maturity $T$ and tranche notional $N_{\mathrm{tr}} = \text{EUR } 10\text{mn}$, assuming homogeneous recovery $\mathrm{Rec} = 40\%$ (hence $\mathrm{LGD} = 60\%$) and $N = 125$ names. Each name has weight $1/125 = 0.8\%$ in the index, so that each default contributes equally

$$\Delta\mathrm{Loss} = \frac{1}{125}\mathrm{LGD} = 0.8\% \times 60\% = 0.48\%$$

to the portfolio loss. After 6 defaults, the loss is given by $\mathrm{Loss} = 6 \times 0.48\% = 2.88\% < 3\%$, hence the $3\% - 6\%$ tranche is still unaffected. At the 7-th default, $\mathrm{Loss} = 7 \times 0.48\% = 3.36\%$ and the attachment point is breached. The tranche loss fraction becomes

$$\frac{3.36\% - 3\%}{6\% - 3\%} = \frac{0.36\%}{3\%} = 12\%,$$

so that the monetary loss on the tranche equals $0.12 \times 10\text{mn} = \text{EUR } 1.2\text{mn}$, paid by the protection seller on the protection leg.

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### 9.3.5 The 1-factor Gauss model

In the following, we consider highly liquid synthetic CDO tranches referencing CDS indices such as the iTraxx (with $N = 125$ constituents). To model the correlated default intensities of the individual consituents of the tranches, we make a drastic simplification considering the 1-factor Gauss copula model assuming a single correlation parameter.

**Definition 9.7** (The 1-factor Gauss copula model.)**.** Let $S \sim \mathcal{N}(0,1)$ denote the systematic factor and $Y_i \sim \mathcal{N}(0,1)$ idiosyncratic factors, mutually independent. Define the latent variables

$$X_i = \rho_i S + \sqrt{1-\rho_i^2}\, Y_i, \qquad i = 1, \dots, N, \tag{122}$$

so that each $X_i \sim \mathcal{N}(0,1)$ and $X = (X_1, \dots, X_N) \sim \mathcal{N}(0, \Sigma)$ with

$$\Sigma_{ij} = \rho_i \rho_j, \quad i \neq j, \qquad \Sigma_{ii} = 1. \tag{123}$$

The *single-parameter one-factor Gaussian copula model* follows from the assumption $\rho_i \equiv \rho \in (-1,1)$ for $i = 1, \dots, N$.

As an additional simplification, let us consider the *Large homogeneous model* (LHM) in which all names share the same (marginal) default intensity $\lambda_i \equiv \lambda$ and the same LGD. Let $\Lambda(t) := \int_0^t \lambda(u)\, du$ (in the time-homogeneous case $\Lambda(t) = \lambda t$), hence

$$\mathbb{Q}(\tau_i < t) = 1 - e^{-\Lambda(t)}. \tag{124}$$

Under these assumption, the present value of the synthetic CDO can be expressed in a concrete form. When conditioning on $S = s$ for some $s \in \mathbb{R}$, it follows from the 1-factor Gauss copula model that

$$X_i | S = s \sim \mathcal{N}\left(\rho s, \sqrt{1-\rho^2}\right) \tag{125}$$

for each $i = 1, \dots, n$. Hence, by definition of the Gauss copula with marginal distributions $\mathbb{Q}(\tau_i < t) = 1 - e^{-\Lambda(t)}$, we have that

$$\Lambda(\tau_i) = -\ln\left(\Phi(X_i) - 1\right) \tag{126}$$

and thus

$$\begin{aligned}
\mathbb{Q}(\tau_i < t | S = s) &= \mathbb{Q}(\Lambda(\tau_i) < \Lambda(t) | S = s) \\
&= \mathbb{Q}\left(X_i < \Phi^{-1}(1 - e^{-\Lambda(t)}) | S = s\right) \\
&= \mathbb{Q}\left(Y_i < \frac{\Phi^{-1}(1 - e^{-\Lambda(t)}) - \rho s}{\sqrt{1-\rho^2}} | S = s\right) \\
&= \Phi\left(\frac{\Phi^{-1}(1 - e^{-\Lambda(t)}) - \rho s}{\sqrt{1-\rho^2}}\right)
\end{aligned} \tag{127}$$

Given $S = s$, defaults are independent and identically distributed. As a consequence, the number of defaults up to $t$ follows a Bernoulli distribution with probability $p_s(t) = \mathbb{Q}(\tau_i < t | S = s)$ so that

$$\mathbb{Q}(\#\{i : \tau_i \leq t\} = n \,|\, S = s) = \binom{N}{n} p_s(t)^n \big(1 - p_s(t)\big)^{N-n}. \tag{128}$$

Consequently, with $\mathrm{Loss}(t) = \mathrm{LGD} \cdot \frac{n}{N}$ in the LHM, the unconditional distribution follows by integrating out $S$:

$$\mathbb{Q}\Big(\mathrm{Loss}(t) = \mathrm{LGD} \cdot \frac{n}{N}\Big) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \binom{N}{n} p_s(t)^n \big(1 - p_s(t)\big)^{N-n} e^{-s^2/2}\, ds. \tag{129}$$

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By the strong law of large numbers, we have

$$\mathbb{E}((\text{Loss}(t) - A)^+ | S = s) \to (\text{LGD}p_s(t) - A)^+ \tag{130}$$

so that

$$\mathbb{E}(L^{A,B}(t)|S = s) \to \frac{1}{B - A}\left[(\text{LGD}p_s(t) - A)^+ - (\text{LGD}p_s(t) - B)^+\right] \tag{131}$$

$$\mathbb{E}(L^{A,B}(t)) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \mathbb{E}(L^{A,B}(t)|S = s)e^{-\frac{1}{2}s^2} ds \tag{132}$$

## 9.4 Counterparty Credit Risk and xVA

### 9.4.1 Expected Loss

The present value of a non-defaultable zero coupon bond is given by

$$P(t, T) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(s)\, ds}\middle|\mathcal{F}_{\sqcup}\right] \tag{133}$$

For a risky zero-bond with recovery payment at maturity, we have

$$\begin{aligned} P(t, T)_d &= \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(s)\, ds}\left(R\mathbb{1}_{\{\tau \le T\}} + \mathbb{1}_{\{\tau > T\}}\right)\middle|\mathcal{G}_{\sqcup}\right] \\ &= P(t, T)(R + (1 - R)S(t, T)) \end{aligned} \tag{134}$$

Hence, the difference between the present values of the non-defaultable and defaultable zero bond, in the case of stochastic independence between interest rates and default intensities, is given by

$$P(t, T) - P(t, T)_d = (1 - R)P(t, T)(1 - S(t, T)) = \text{LGD} \cdot P(t, T) \cdot \text{PD}([t, T]) \tag{135}$$

$$EE(t) = \mathbb{E}(V()\mathbb{E}^{\mathbb{Q}}(e^{-\int_t^T r(s)\, ds}|\mathcal{F}_t) \tag{136}$$

### 9.4.2 The Credit Value Adjustment (CVA)

The Credit Value Adjustment (CVA) is defined as the averaged discounted expected loss

$$\text{CVA}(t) = \mathbb{E}\left[D(t, \tau) \cdot \text{LGD}(\tau) \cdot V_\tau^+\right] \tag{137}$$

with $V_\tau^+$ the positive future value of the portfolio at time of default $\tau$. Assuming a constant recovery rate $R$, this yields

$$\text{CVA}(t) = (1 - R)\left[D(t, \tau) \cdot V_\tau^+\right] \tag{138}$$

$$= (1 - R)\int_t^T \mathbb{E}\left[D(t, s)V_t^+ \mathrm{d}\mathbb{P}(\tau \in (s, s + ds])\right] ds \tag{139}$$

$$= (1 - R)\int_t^T \mathbb{E}\left[D(t, s)V_t^+ \mathbb{1}_{\{\tau \in [t, t+dt]\}}\right] ds \tag{140}$$

$$(1 - R)\int_0^T \mathbb{E}\left[V_t^+ \mathbb{E}\left[\mathbb{1}_{\{\tau \in [t, t+dt]\}}|\mathcal{F}_t\right]\right] dt \tag{141}$$

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$$\text{CVA} = (1-R)\int_0^T \mathbb{E}\left[V_t^+ \mathbb{1}_{\{\tau\in[t,t+dt]\}}\right] dt \tag{142}$$

$$(1-R)\int_0^T \mathbb{E}\left[V_t^+ \mathbb{E}\left[\mathbb{1}_{\{\tau\in[t,t+dt]\}}|\mathcal{F}_t\right]\right] dt \tag{143}$$

## 9.5 Counterparty Credit Risk and xVA

### 9.5.1 Expected Loss

The present value of a non-defaultable zero coupon bond is given by

$$P(t,T) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(s)\,ds}\middle|\mathcal{F}_t\right]. \tag{144}$$

For a risky zero bond with recovery of face value $R$ paid at maturity, we obtain

$$\begin{aligned} P_d(t,T) &= \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(s)\,ds}\left(R\mathbf{1}_{\{\tau\le T\}} + \mathbf{1}_{\{\tau>T\}}\right)\middle|\mathcal{G}_t\right] \\ &= P(t,T)\left(R + (1-R)S(t,T)\right), \end{aligned} \tag{145}$$

where $S(t,T) = \mathbb{Q}(\tau > T|\mathcal{G}_t)$ denotes the survival probability.

Assuming stochastic independence between interest rates and default,

$$P(t,T) - P_d(t,T) = (1-R)P(t,T)(1-S(t,T)) \tag{146}$$

$$= \text{LGD}\cdot P(t,T)\cdot \text{PD}([t,T]), \tag{147}$$

where $\text{LGD} = 1-R$ and $\text{PD}([t,T]) = 1 - S(t,T)$.

For a general portfolio with future value $V_t$, the (risk-neutral) discounted expected exposure at time $t$ is

$$EE(t) = \mathbb{E}^{\mathbb{Q}}\left[D(0,t)\,V_t^+\right], \tag{148}$$

where $V_t^+ = \max(V_t, 0)$ and

$$D(0,t) = \exp\left(-\int_0^t r(s)\,ds\right).$$

### 9.5.2 The Credit Value Adjustment (CVA)

The Credit Value Adjustment (CVA) is defined as the discounted expected loss due to counterparty default:

$$\text{CVA}(t) = \mathbb{E}^{\mathbb{Q}}\left[D(t,\tau)\,\text{LGD}(\tau)\,V_\tau^+\,\middle|\,\mathcal{G}_t\right], \tag{149}$$

where

$$D(t,u) = \exp\left(-\int_t^u r_s ds\right)$$

and $\mathcal{G}_t$ denotes the enlarged filtration including default information.

Assuming constant recovery $R$ and $\text{LGD} = 1-R$, we obtain

$$\text{CVA}(t) = (1-R)\mathbb{E}^{\mathbb{Q}}\left[D(t,\tau)\,V_\tau^+\,\middle|\,\mathcal{G}_t\right]. \tag{150}$$

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**General representation (no independence assumption)** Using the default indicator measure,

$$\mathrm{CVA}(t) = (1-R)\mathbb{E}^{\mathbb{Q}}\left[\int_t^T D(t,s)\, V_s^+ \mathbf{1}_{\{\tau\in ds\}} \,\middle|\, \mathcal{G}_t\right]. \tag{151}$$

Applying the tower property and conditioning on $\mathcal{F}_s$ yields

$$\mathrm{CVA}(t) = (1-R)\int_t^T \mathbb{E}^{\mathbb{Q}}\left[D(t,s)\, V_s^+ \mathbb{E}^{\mathbb{Q}}\left[\mathbf{1}_{\{\tau\in ds\}} \mid \mathcal{F}_s\right] \,\middle|\, \mathcal{G}_t\right]. \tag{152}$$

Up to this point, no independence between exposure, interest rates or default has been assumed.

### Intensity-Based Model

Let $\lambda_t$ be the $\mathcal{F}_t$-adapted default intensity. The conditional survival probability is

$$\mathbb{Q}(\tau > u \mid \mathcal{F}_t) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^u \lambda_s ds\right) \,\middle|\, \mathcal{F}_t\right]. \tag{153}$$

In particular,

$$S(t,u) := \mathbb{Q}(\tau > u \mid \tau > t, \mathcal{F}_t) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^u \lambda_s ds\right) \,\middle|\, \mathcal{F}_t\right]. \tag{154}$$

The conditional default probability over $[s, s+ds]$ is

$$\mathbb{E}^{\mathbb{Q}}\left[\mathbf{1}_{\{\tau\in[s,s+ds]\}} \mid \mathcal{F}_s\right] = \mathbb{Q}(\tau > s \mid \mathcal{F}_s) - \mathbb{Q}(\tau > s+ds \mid \mathcal{F}_s) \tag{155}$$

$$= \mathbb{Q}(\tau > s \mid \mathcal{F}_s)\,(1 - S(s, s+ds)). \tag{156}$$

Since the default intensity process $(\lambda_t)_t$ is $\mathcal{F}_t$-measurable, we have

$$\begin{aligned}\mathbb{Q}(\tau > t+\Delta t|\mathcal{F}_t) &= \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_0^t \lambda_s ds\right)\exp\left(-\int_t^{\Delta t} \lambda_s ds\right) \,\middle|\, \mathcal{F}_t\right] \\ &= \exp\left(-\int_0^t \lambda_s ds\right)\mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^{\Delta t} \lambda_s ds\right) \,\middle|\, \mathcal{F}_t\right]\end{aligned} \tag{157}$$

Hence the CVA admits the survival-probability representation

$$\mathrm{CVA}(t) = (1-R)\int_t^T \mathbb{E}^{\mathbb{Q}}\left[D(t,s)\, V_s^+ \mathbb{Q}(\tau > s \mid \mathcal{F}_s)\,(1 - S(s, s+ds))\right]. \tag{158}$$

This representation is fully general and does not require independence between exposure, interest rates and default intensity.

On a time grid $t = t_0 < \cdots < t_N = T$,

$$\mathrm{CVA} \approx (1-R)\sum_i^N \mathbb{E}\left[D(t,t_i)V_{t_i}^+ \exp\left(-\int_t^{t_i} \lambda_s ds\right)(1 - S(t_i, t_i+\Delta t))\right] \tag{159}$$

This is the discrete formula as implemented in the Monte Carlo risk engine.

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